Aggregate history, hidden lane allocation, and useful common advice
Conditional analytical results and the remaining obstruction

Freeway bottleneck review continuation

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Status. Methodology only. These are original conditional derivations, not simulations, an adoption of a traffic law, or a proof about native SUMO. The approved target is an admissibly prepared pair with equal strong aggregate history and different useful shared desired-speed-advice decisions. We obtain an information-side result in a free-flow model class and a decision-side negative result in a fixed-service queue. Neither completes that target.

1 Information-side construction without deleting history

Consider three upstream lanes, one of which ends at a fixed downstream lane drop. Before any traffic reaches a restrictive boundary, let $k_\ell(t,x)$ denote continuous density in mass/metre and let all lanes have the same free-flow speed $v_0 > 0$ in metres/second. Let $r_{\ell j} \geq 0$ be conservative lateral transfer rates in mass/(metre second), with transfers paired on the same longitudinal coordinate. The conditional free-flow equations are

$$\partial_t k_\ell + \partial_x(v_0 k_\ell) = \sum_{j\neq\ell} r_{j\ell} - \sum_{j\neq\ell} r_{\ell j}. \tag{1}$$

We require admissible, nonnegative states, adequate receiving/storage, and an identical homogeneous free-flow branch throughout the prefix. Transfers need not be zero, symmetric, or independent of density. We do not clip or repair a published transfer law to enforce these assumptions.

Let $K = \sum_\ell k_\ell$, and provide the controller the entire predecision history of spatial mass and density, mass-weighted mean speed, empty-bin masks, cumulative aggregate flows across spatial boundaries, source waiting and input (both intended and actual), together with common geometry, response parameters, advice and future forcing. Lane-entry allocation is the sole unknown past input in this construction. There are no connectors occupied during this prefix.

Proposition 1 (Exact aggregate-prefix invariance)**.** *Starting empty with identical total entry rate $e(t)$ and identical aggregate boundary conditions, two admissible solutions of* (1) *have identical declared aggregate histories, regardless of their hidden lane-entry allocations. This holds on any spatial/time sampling of these observables.*

Proof. Pairwise lateral terms cancel, giving

$$\partial_t K + v_0 \partial_x K = 0, \qquad K(0,x) = 0, \qquad v_0 K(t,0) = e(t). \tag{2}$$

The transport solution is determined by the same total input and is therefore identical. Every spatial-bin mass is its integral, every aggregate crossing flow is $v_0 K$, and mean speed in every occupied bin is v_0. Empty bins have the same mask and the same declared empty-speed convention. Choose entry below the receiving limits of either allocation, so actual and intended entry agree and source waiting is zero in both. The other known inputs are identical. No current-state-only reduction or shorter history has been used. □

A reachable hidden contrast in the conditional class

The result alone does not guarantee that a lane contrast survives. A short-prefix example can establish that separately. Suppose, additionally, total outgoing lateral transfer from any lane is bounded by λ times its current mass, where $\lambda < \infty$ is in inverse seconds. This is a hypothesis about a valid transfer law, not permission to cap an invalid one. Inject the same low total mass m over $[0, T]$: preparation a through the ending lane and b through a continuing lane. Let no traffic reach the lane-drop boundary or leave the upstream region during this short prefix. Total mass satisfies $M(t) \le m$. For ending-lane upstream mass E_a, E_b, the transfer bound gives

$$E_a(T) \ge m - \lambda \int_0^T M(t)\, dt \ge m - \lambda m T, \tag{3}$$

$$E_b(T) \le \lambda \int_0^T M(t)\, dt \le \lambda m T, \tag{4}$$

$$E_a(T) - E_b(T) \ge m(1 - 2\lambda T) > 0 \quad \text{if } T < (2\lambda)^{-1}. \tag{5}$$

For $\lambda = 0$, any sufficiently short prefix works. Small enough entry rate keeps the states on the declared free-flow branch. Thus known-empty initialization and full aggregate history do not, by themselves, rule out a reachable lane contrast when lane-entry forcing is hidden.

Scope of this witness. The contrast is upstream ending-lane mass at an early decision, not certified contrast in the last 400 metres of the approved numerical design. Persistence to that region and postdecision control value remain unproved. These are continuous-mass observation analogues, not bit-identical native vehicle-count/trajectory arrays. If lane inputs are also known, this construction no longer establishes unknown-input ambiguity. If actual free-flow speed depends on lane/gaps or receiving limits activate, Proposition 1's premises must be checked again; they cannot be assumed from homogeneous vehicle types.

Traffic-source grounding and non-transfer

FB0030 gives paired lane conservation and triangular demand in its equations 1–2 and 12–15. In an admissible, demand-active homogeneous free-flow branch, the paired modifications in equations 20–22 also cancel upon summation over lanes. This motivates the model class above, but does not certify the complete published discrete scheme. Its calibrated lanes have different fundamental diagrams; empty-density ratios, transfer budgets and receiving/capacity branches remain unresolved. We have not transplanted the continuous short-prefix witness into that scheme, disabled mandatory/courtesy incentives, or proven its lane contrast persists. See FB0030.

2 Why a hidden contrast is not yet useful advice information

We now test the tempting completion: attach a fixed bottleneck capacity to each hidden preparation, and claim this alone makes slowing advice differently useful. It does not in the following explicitly restricted class.

For each hidden state s, model the bottleneck as a work-conserving fluid point queue, initially empty, with nonnegative locally integrable service rate $c_s(t)$. The rate may differ between preparations, but is fixed independently of the subsequent advice. Let $B_{s,u}(t)$ be cumulative arrivals to that queue under common desired-speed advice trajectory u. Let $S_s(t) = \int_0^t c_s(z)\, dz$. Assume some common feasible trajectory u^+ gives arrival dominance:

$$B_{s,u^+}(t) \ge B_{s,u}(t) \quad \text{for all } s, u, t. \tag{6}$$

For an independent free-flow approach with monotone desired-speed response, holding the highest feasible advice supplies such a trajectory. This assumption is *not* inferred for native following, lane changing or spillback.

Queue departures satisfy the continuous min-plus expression

$$D_{s,u}(t) = \inf_{0 \le r \le t} \{B_{s,u}(r) + S_s(t) - S_s(r)\}, \qquad B_{s,u}(0) = 0. \tag{7}$$

Use continuous cumulative arrivals here; atomic time-zero batches require an explicit initial-queue convention and are not needed for this derivation. Let $I(t)$ be cumulative intended departures for a common finite cohort of mass m, including traffic still waiting at source or on the approach. Define the complete all-cohort time-spent score through queue exit by

$$J_s(u) = \frac{1}{m} \int_0^\infty [I(t) - D_{s,u}(t)]\, dt. \tag{8}$$

Assume every cohort drains and scores are finite; a common fixed downstream travel time adds an action-independent constant. Any already accrued prefix cost is likewise fixed within each preparation.

Proposition 2 (A common best action in the fixed-service class)**.** *Under these assumptions, $J_s(u^+) \le J_s(u)$ for every state and feasible advice. Consequently no pair in this class requires disjoint optimal or nonnegative- tolerance acceptable action sets over the full feasible advice space.*

Proof. For the same S_s, arrival dominance makes every expression inside the infimum in (7) no smaller for u^+. Thus $D_{s,u^+}(t) \ge D_{s,u}(t)$ pointwise. The intended-input process in (8) is unchanged, so integration gives the score inequality. This argument counts delayed arrivals on the approach/source; moving waiting outside the bottleneck cannot create an apparent improvement. It holds separately for every s, including different fixed capacities. All states share u^+. □

For a feasible set $\mathcal{U}$ containing u^+ and acceptable tolerance $\varepsilon_s \ge 0$, this means

$$\mathcal{A}_s = \{u \in \mathcal{U} : J_s(u) - \inf_{v \in \mathcal{U}} J_s(v) \le \varepsilon_s\}, \qquad u^+ \in \bigcap_s \mathcal{A}_s, \tag{9}$$

$$\inf_{u \in \mathcal{U}} \max_s \left[J_s(u) - \inf_{v \in \mathcal{U}} J_s(v) \right] = 0. \tag{10}$$

This is a conditional obstruction, not proof that real common speed advice cannot help. A mechanism that violates arrival dominance or makes realized service depend on advice/state evolution can escape it. Capacity drop is one possible mechanism, not a necessary condition for every traffic-control benefit.

3 The missing mechanism and the action-menu qualification

The remaining target needs a jointly justified chain: admissible lane/history collision near the consequential region; shared desired-speed response; endogenous merge/receiving dynamics under that response; and disjoint useful-action sets after all slowing/source/recovery costs. Positive lane effects under another actuator do not complete this chain.

FB0001 provides an aggregate speed-limit/capacity-drop analytical example, but its equations 5/9 use a discontinuous discharge switch and equation 6 regulates upstream inflow rather than the approved desired-response zone. FB0038 optimizes lateral fractions with natural lane changing turned off; it is not shared advice. FB0039 supplies field plausibility for common speed limits affecting lane distribution, not a dynamic voluntary-advice closure or paired-history certificate. See FB0001, FB0038, and FB0039.

The prospective nine-action menu has baseline advice 90 km/h and early/late 60-second pulses of $\pm 5, \pm 10$, reverting to 90. It does not contain holding 100 km/h throughout the decision

horizon. Proposition 2 is therefore a result for a feasible advice space containing the dominating action, not a negative result for that pulse menu. Restricted-menu distinctions could still occur without unavoidable continuous-space information loss. Conversely, including held 100 in a native reference would not assume it is actually best there. An added dominance stress probe requires explicit menu/budget approval, not silent expansion. With the existing maximum paired/context/time-step counts, one added action changes action invocations from 216 to 240 and the total cap from 480 to 504; preparation counts and other obligations remain unchanged.

4 Certificate disposition

- Established conditionally: exact full aggregate-prefix equality and a reachable early lane contrast in a homogeneous conservative free-flow class.
- Established conditionally: fixed, advice-independent work-conserving service plus arrival dominance cannot produce an unavoidable useful-action split.
- Not established: the near-drop preparation, native full-history collision, an admissible shared-advice/merge service closure, or a useful-action split.
- Not established: minimum sufficient retained state, any gradient correctness, independent gradient usefulness, field relevance, or generic aggregate sufficiency.

We stop at the evidence-route decision. No unsupported lane penalty, shortened history, categorical moving-mass process, smoothing or saltation exception has been adopted to bridge the missing link.